

Competitor Product Report

Newest products, content, and market positioning across radiosonde manufacturers

Generated: 2026-06-12T00:00:00Z

Prepared for Graw Radiosondes GmbH

Companies covered:

- Vaisala
- Meisei Electric
- InterMet Systems
- Meteomodem
- Lockheed Martin (LMS6 / Sippican)
- Meteolabor AG

RS41 / RS41 E-models

product

Vaisala's flagship upper-air radiosonde line. RS41 E-models add BioCover and BioTwine (patent-pending biodegradable, plastic-free cover/insulation and unwinder twine), cutting plastics by 66% with the same performance as standard RS41. Standard line includes RS41-SG, RS41-SGP, RS41-SGM variants with different sensor/transmission configs.

Use cases:

- Operational synoptic upper-air soundings
- Numerical weather prediction data input
- Ozone sounding (with ECC sensors)
- Sustainability-focused sounding programs (biodegradable materials)

Target industries:

National meteorological services, Research institutions/GRUAN network, Airports

Note: RS41-SGM model specifically marketed with 'radio silence and encryption' for improved security - relevant to defense-adjacent buyers. RS41 is described as the most-used GRUAN reference-grade radiosonde.

<https://www.vaisala.com/en/products/weather-environmental-sensors/upper-air-radiosondes-rs41-rs41-e-models>

Cirrus Sounding System MW51 (Multi-GNSS / message authentication upgrade)

product/feature

Upgrade path that unlocks Multi-GNSS support (uses GPS, Galileo, BeiDou etc.) and industry-first message authentication (MAC) for RS41 radiosondes, improving resilience to GPS interference by ~60% and protecting against data tampering in transit.

Use cases:

- Defending against GPS jamming/interference (hybrid threats)
- Ensuring data integrity/authenticity of weather messages

Target industries:

National meteorological services, Defense / military meteorology, Critical infrastructure operators

Note: Explicitly framed around 'hybrid threats' and cybersecurity for meteorological agencies - a distinct marketing angle Graw should be aware of.

<https://www.vaisala.com/en/products/weather-environmental-sensors/upper-air-radiosondes-rs41-rs41-e-models>

AWS810 Airport Edition

product (newly launched, adjacent)

Purpose-built automatic weather station for continuous, ICAO/WMO-compliant airport weather observations, integrating with AviMet systems, scalable from a single runway to a full airfield. Announced May 26, 2026.

Use cases:

- Airport weather observation and compliance reporting

Target industries:

Airports / aviation

Date: 2026-05-26

Note: Not a radiosonde, but shows Vaisala expanding adjacent weather-sensing product lines for the airport/aviation vertical - useful market-context signal even if outside core radiosonde scope.

<https://www.vaisala.com/en/news-and-insights>

RM60 precipitation sensor

product (newly launched, adjacent)

Described as the world's first truly maintenance-free precipitation sensor, addressing chronic under-measurement of precipitation and unmonitored locations. Announced May 5, 2026.

Use cases:

- Precipitation network monitoring

Target industries:

National meteorological services, Climate/precipitation monitoring networks

Date: 2026-05-05

Note: Adjacent product (not a radiosonde) but indicates R&D; investment direction. Most of Vaisala's news feed is investor/financial (Manager Transactions, Share Repurchases) - filtered out as not relevant.

<https://www.vaisala.com/en/news-and-insights>

iMS-100 eco GPS Radiosonde

product

Compact, lightweight (38g) GPS radiosonde measuring wind speed/direction, pressure, temperature, and humidity. Reduces operational costs (balloon, gas, shipping) and improves safety on ground impact. Optional bio-material packaging reduces environmental burden.

Use cases:

- Routine upper-air soundings with reduced cost/weight
- Environmentally-conscious sounding programs

Target industries:

National meteorological services, Weather observation networks (e.g. AMeDAS-type)

Note: Direct weight/sustainability competitor to Graw's lightweight offerings and to Meteomodem M20/Vaisala RS41-E.

https://www.meisei.co.jp/english/products/products_category/meteorology

RS-11G GPS Radiosonde

product

GPS radiosonde with improved temperature sensor accuracy via repositioning and a high-response humidity sensor. Very light at 85g including batteries, improving safety if it falls on land.

Use cases:

- Operational upper-air soundings
- High-accuracy temperature/humidity measurement

Target industries:

National meteorological services

https://www.meisei.co.jp/english/products/products_category/meteorology

RS-06G GPS Radiosonde

product

GPS radiosonde offering a 240-minute sounding timeframe on two lithium batteries, covering the full flight from pre-flight through balloon burst.

Use cases:

- Extended-duration soundings

Target industries:

National meteorological services, Research institutions

https://www.meisei.co.jp/english/products/products_category/meteorology

Automated Radiosonde System (ARS)

product

Canister-type automated launcher that adjusts the number of radiosondes/balloons loaded, removing the need for manual mid-night/early-morning launches and hazardous H2 gas inflation work.

Use cases:

- Unattended/automated upper-air sounding stations

Target industries:

National meteorological services (remote/automated stations)

Note: Comparable to Meteomodem's Robotsonde automated launcher concept.

https://www.meisei.co.jp/english/products/products_category/meteorology

CPS sonde (Cloud Particle Sensor) / HYVIS

product

Add-on sondes that directly measure atmospheric airborne particles (cloud particles, volcanic ash) in combination with RS-11G; HYVIS is a balloon-borne instrument capturing upper-air cloud/ice crystal particle images.

Use cases:

- Cloud physics research
- Volcanic ash monitoring

Target industries:

Research institutions, Volcanology/aviation safety monitoring

Note: Niche scientific accessory sondes - shows Meisei's breadth beyond core PTU radiosondes.

https://www.meisei.co.jp/english/products/products_category/meteorology

InterMet Systems

CopterSonde / CopterSonde 3

product (UAV platform)

Drone-based (UAV) atmospheric sensing platform developed with University of Oklahoma research, used to study the atmospheric/marine boundary layer, hurricanes, and convective storms. Featured by the Wall Street Journal and NPR; ongoing CRADA with NOAA's National Severe Storms Laboratory; tied to a \$1.4M OU wildfire-research grant.

Use cases:

- Boundary-layer atmospheric profiling
- Severe storm / hurricane research
- Wildfire research and mitigation

Target industries:

Research institutions / universities, Government weather research labs (NOAA), Wildfire management agencies

Date: 2025-05-12

Note: This is InterMet's most actively promoted current product/initiative - significant UAV pivot away from traditional balloon-borne radiosondes. High-value finding for the 'newest products' report.

<https://www.intermetsystems.com/news/>

iMet-4 Radiosonde / iMet-4 RSB Research Radiosonde

product

InterMet's core balloon-borne radiosonde line for research and operational upper-air sounding.

Use cases:

- Operational and research upper-air soundings

Target industries:

National weather services, Universities / research institutions

<https://www.intermetsystems.com/products/radiosondes/>

iMet-1500M 1680 MHz Military Portable Sounding System

product

1680 MHz GPS/RDF sounding system marketed specifically as a military-portable configuration.

Use cases:

- Field/portable military meteorological soundings

Target industries:

Military / defense

Note: Explicit military product line - InterMet segments military vs civilian (403 MHz synoptic vs 1680 MHz military) sounding systems.

<https://www.intermetsystems.com>

Meteomodem

M20

product

Ultra-light (36g) radiosonde, usable without a parachute, saving ~20% gas. Heated humidity sensor reduces icing/saturation issues. Compatible with Robotsonde automated launcher. Marketed as the lightest and greenest radiosonde on the market. Recently used in a campaign with small 50g balloons reaching 15km altitude.

Use cases:

- Cost- and gas-efficient routine soundings
- Automated/unattended launches (via Robotsonde)
- Low-cost high-altitude soundings with small balloons

Target industries:

National meteorological services, Operational meteorology programs focused on sustainability/cost

Note: Includes cybersecurity features: external ON/OFF power switch and release authorization indicated on the radiosonde itself.

<https://www.meteomodem.com/m20>

M10

product

World-reference radiosonde (95x95x88.5mm, 150g incl. batteries) used by national meteorological services, military forces, and research institutes. Single low-power electronic board, 4h battery life, Robotsonde-compatible, GNSS-based pressure, resistant to GNSS spoofing, GRUAN certification in progress.

Use cases:

- Meteorological services: routine upper-air radiosounding
- Military forces: meteorological/aeronautical information and ballistic tests
- Research: laboratories, institutes, universities (GUAN/GRUAN sites)

Target industries:

National meteorological services, Military / defense, Research institutions

Note: Explicitly lists 'Military forces' and 'ballistic tests' as a target use case/industry - directly comparable to Graw's defense-relevant lines. 'Resistant to GNSS Spoofing' is a security selling point similar to Vaisala's multi-GNSS/MAC features.

<https://www.meteomodem.com/m10>

SR20 receiver

product (newly launched)

New Meteomodem ground receiver, described as more powerful and more robust, designed to meet the sector's most demanding requirements, fully compatible with the entire Meteomodem range, for mobile or stationary operations.

Use cases:

- Ground reception/processing for mobile or fixed sounding stations

Target industries:

National meteorological services, Military / mobile field operations

Note: Featured prominently on the homepage as a recent/new release - candidate highlight for the newest-products report. No specific launch date found on-page.

<https://www.meteomodem.com>

Pilotsonde (PS20)

product

GNSS-based pilot sonde (no temperature/humidity sensors) for high-altitude wind measurement, lightest on the market at 32g, alternative to theodolite systems. Compatible with SR10 ground station, color-coded (red/green) for easy frequency identification, optional IR remote for frequency change. Developed since 2013.

Use cases:

- High-altitude wind profiling (pilot balloon replacement)
- Military forces: ballistic tests, aeronautical weather info, skydiving pre-flight checks
- Aeronautic recreational activities (hot-air balloon pre-flight tests)

Target industries:

National hydro-meteorological services (NHMSS), Military / defense, Aviation / recreational aeronautics

Note: Broad industry targeting explicitly listed on-page (NHMSS, military, aeronautic recreational) - good example of multi-industry positioning for Graw to benchmark against.

<https://www.meteomodem.com/pilotsonde>

Lockheed Martin (LMS6 / Sippican)

LMS6 Radiosonde + LMG6 Ground System

product

Latest Lockheed Martin radiosonde model using current GPS receivers and advanced temperature/humidity sensing. Factory-calibrated (no preflight baseline checks needed), lithium battery power for simplified prep and extended shelf life, ETSI-compliant transmitters with digital error correction. Over 6 million radiosondes provided globally over 60+ years (incl. legacy Sippican products).

Use cases:

- High-resolution atmospheric profiling to >30km altitude

Target industries:

Meteorological organizations, Research institutions, Military agencies

Note: Lockheed's general news page ([lockheedmartin.com/en-us/news.html](https://www.lockheedmartin.com/en-us/news.html)) is entirely broad defense/aerospace corporate news with no meteorological-specific items - confirms it's not a useful monitoring source for product updates; the dedicated meteorological-instrumentation product page is the only useful source found.

<https://www.lockheedmartin.com/en-us/products/meteorological-instrumentation.html>

SRS-C50 digital radiosonde

product

Newly developed digital radiosonde that can be equipped with different sensors, measuring temperature, humidity, air pressure, ozone and wind.

Use cases:

- Configurable multi-parameter atmospheric sounding (incl. ozone)

Target industries:

Meteorological research, Specialized sounding programs (e.g. ozone monitoring)

Note: Niche/small-scale player; described as newly developed but no specific date available.

<https://www.meteolabor.ch/meteorologie?lang=en>

Argus 48 system

product/solution (custom)

Custom system developed for the Swiss Air Force consisting of radiosondes plus a mobile ground station, used to probe the exact weather situation on-site.

Use cases:

- Military field weather reconnaissance (mobile ground station + radiosondes)

Target industries:

Military / defense (Swiss Air Force)

Note: Direct evidence of a competitor's defense customer (Swiss Air Force) and a custom mobile radiosonde+ground-station bundle - directly relevant to Graw's defense-market positioning.

<https://www.meteolabor.ch>